

GROUP 3

Heart Disease Risk Score

Track	Machine Learning
Duration	2 Weeks (10 working days, 4 hours/day)
ML Task	Classification
Classification Type	Binary Classification
Target	Heart-disease risk class
User Interface	Required

Project Analysis

This educational project uses health-related input variables to build a machine-learning model that estimates a heart-disease risk class. Students must focus on the ML workflow, model comparison, evaluation, and responsible presentation of results.

The application must be presented as an educational risk-prediction project, not as a medical diagnosis system.

What the Student Must Build (Mandatory)

- Load and inspect the approved heart-disease dataset.
- Check missing values, duplicates, data types, and class balance.
- Perform EDA.
- Prepare features and the target variable.
- Split into training and testing data.
- Train at least two classification models.
- Evaluate using appropriate metrics and a confusion matrix.
- Compare models and select the best one.
- Save and reload the final model.
- Build a functional input form and result page.
- Display the predicted risk class clearly.

Technical Requirements

- Python
- Pandas
- NumPy
- Scikit-learn
- Joblib/Pickle
- Git & GitHub
- Streamlit/Gradio or another suitable UI framework

Dataset Requirements

- Use the larger instructor-approved heart-disease dataset provided in the repository.

- Document the exact target column and class meaning from the supplied file.
- Typical fields may include age, sex, chest-pain type, resting blood pressure, cholesterol, fasting blood sugar, ECG features, maximum heart rate, exercise-induced angina, oldpeak, slope, ca, and thal.

Important Note

Educational use only. The application must not claim to diagnose heart disease or replace a healthcare professional.

Optional Features (Bonus)

These features may improve the project but are not required for basic project completion.

- Risk score visualization
- Feature importance
- Health information summary
- Prediction history

References and Starting Resources

- UCI Heart Disease resources
- Kaggle Heart Disease datasets
- Scikit-learn Documentation